

Will The Apple Fall? A VR Gravity Calculator Through Controller Movement Tracking and Gaze Interaction

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Figure 1: Capture of the gravitational force system: (a) The user inputs mass and radius through controller rotation tracking; apples float around based on the gravity output; (b)(c) The user selects material presets of marshmallow through hand-eye coordination.

ABSTRACT

We introduce a VR gravity calculator for the "Next-Gen Calculators" challenge. This interactive tool enables users to calculate gravitational forces by inputting planetary radius and mass, employing advanced controller movement tracking and eye-hand coordination for interaction. It also has different planets and materials presets, allowing automatic data retrieval and hypothetical exploration. It visualises and simulates the gravitational forces in real-time through a fun fruit-falling effect. Our calculator aims to transform geophysics education by making complex concepts accessible and engaging through tangible simulation. This innovative approach bridges theoretical knowledge with practical application, offering a responsive and interactive learning experience in geophysics.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

As one of the fundamental physics concepts, gravity is all around us, from the spinning earth to the falling apple. Understanding gravity's influence is crucial, yet its vast scale and the relative positions of planets present a significant challenge in demonstrating gravitational interactions on planetary surfaces. In response to this challenge, for the 3D UI contest of calculator design, we focus on exploring planet gravitation through an interactive VR experience.

We present a VR gravity calculator, which computes and visualises the gravitational forces exerted by planets. Moreover, this

tool integrates advanced interaction techniques, such as precise hand rotation tracking and eye-hand coordination, with an engaging interface to create an intuitive and immersive educational experience for learners at all levels.

2 VR CALCULATOR

2.1 Gravity System

We created a customised planet gravity system that first calculates the gravitational force and then uses it to simulate objects' interactions around the planet's surface. We scripted the force calculation based on Newton's law of gravitation $F = \frac{GM^1}{R^2}$. The VR software was developed in Unity 2021.3.33f1, and the physics simulation was adapted from an open-source gravity simulation [1].

2.2 User Journey

Users start their journey in a fictional universe scene, featuring a customisable planet object. For the primary calculator function, they can input the numeric values of planetary radius and mass using an innovative interface that tracks the precise movement of the controller. This interface converts the controller's rotational movements and depth into numeric input variables. Alternatively, specific planets or material presets can be selected to obtain predetermined input values through eye gaze selection. After completing the input process, users can confirm their inputs to initiate the calculation. The system then automatically generates a real-time simulation based on the calculated gravitational effect, such as apples floating on the planet's surface. The VR software is engineered for compatibility across multiple platforms.

2.3 Features and Design

There are three main use cases for this VR gravity calculator.

¹Gravitational Constant $G = 6.673 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$.

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- **Customised Value Input.** This function enables users to input planetary radius (in meters) and mass (in kilograms) using innovative rotation tracking of the hand controller to calculate gravitational force (in m/s^2).
- **Planet Presets.** Users can select a planet from the solar system (e.g., Earth, Moon, Jupiter, Mars) through gaze selection to automatically retrieve its radius and mass for gravity calculations and simulations. This feature incorporates pre-configured data based on NASA's planetary facts [4]. We also use the planetary diameter to size the object to visualise the relative size.
- **Material Presets.** This unique aspect of the VR calculator allows users to experiment with hypothetical planet materials and densities, such as "What if Earth were made of marshmallows, gold, or water?" It demonstrates the impact of planetary density on gravity in a fun and engaging way, offering intuitive insights that are challenging to achieve in real-world settings.

3 INTERACTION TECHNIQUES

3.1 Rotational Input System

We developed a movement-based input system to leverage the recorded controller movement data, presenting a novel alternative to traditional button-based calculator UIs. This system transforms the rotational movement of VR controllers into input values. Users initiate the process by pressing and holding the trigger button on the controller. Subsequently, they can rotate the controller to enter precise values while continuing to hold the trigger, as the rotational movement is mapped to the numeric input. Releasing the trigger confirms the selected input.

We choose the controller-rotational-movement technique over other methods (e.g., gesture-based motions) for better precision and usability. Micro-gestures tend to lose precision with arm extension [3], but our system's straightforward wrist twist overcomes this issue, reducing strain on elbows, arms, and torso. This efficient motion lessens physical exertion and fatigue, increasing user comfort as it doesn't necessitate holding the controller in elevated or visible positions [5]. Once users become accustomed to the system, they can position their arms comfortably. Additionally, this rotational method is highly adaptable. Beyond numerical values, it can be easily extended to manage various inputs, such as vectors, planes and polygons, for scientific calculation and computation.

3.2 Depth-Positioned Selection

While rotational movements are precise for inputting values, our testing revealed the need for an additional interactive element. For instance, inputting larger numerical values could require excessive rotation. To address this, we incorporated the controller's forward position to select the adjusted numerical place value.

As the user extends the controller to a specific depth, the corresponding numerical place (ones, tens, thousands) becomes selectable, allowing precise control over the variable's value (see fig.2). This method is motion-based and remains functional even when the controller is not within the user's line of sight.

3.3 Eye-hand Coordination

We integrate eye tracking for the natural VR interaction so that people get the information of what they are looking at [2]. Instead of the traditional controller selection, user can use their gaze ray as a cursor, browse the objects of interest and select by pressing the trigger while fixating on the object selections. This design is available for using the planet and material presets to calculate gravity.

This interaction feature is currently implemented with the Meta Quest Pro using its integrated eye tracker and the Meta Movement

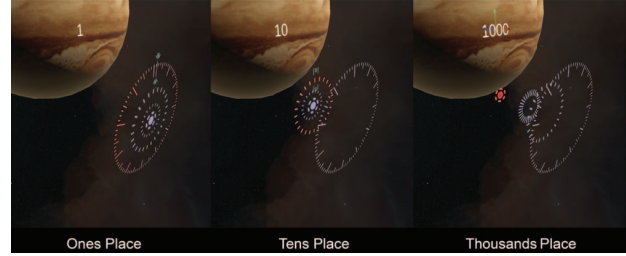


Figure 2: Depth-Positioned Selection. The controller's movement along the "Z" axis for input adjustments across numerical places(ones, tens, or thousands).

SDK. The selection of objects, such as the planet and material presets, were configured to detect the eye gazes following our previous practice by using the ray cast physics for collier detection [6].

4 DISCUSSION & CONCLUSION

We introduce an intuitive VR interface to calculate and simulate gravitational forces on planetary surfaces. Our system features a controller movement tracking system for variable input, capturing the rotational movement of the hand controller and translating it into numerical values. Additionally, we have incorporated forward-depth position tracking to facilitate easier adjustments across numerical places, particularly beneficial when dealing with large numbers. Eye tracking is employed for gaze selection interaction.

A significant advantage of our UI is its operational effectiveness even when the input interface is not centred in the user's view, enhancing usability and freedom to explore the virtual environment. Furthermore, integrating eye tracking deepens user engagement, encouraging the exploration of various presets and fostering a comprehensive understanding of celestial scales and material properties. While this project primarily addresses the calculation and entry of numerical values for gravitational forces, the versatility of our system and UI extends to any application requiring precise and quantifiable inputs across a range of units.

"Will the apple fall?" Our approach to STEM education invites learners to pose and test various hypothetical scenarios in an interactive setting. With this gravity tool, we aim to illustrate how the concept of gravitational force can be taught in a manner that is both intuitive and engaging to learners at all levels.

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